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Date: 21/6/2026

1. Problem Identification and Requirements

Objective

Design an Industrial Internet of Things (IIoT) system that monitors machine health, predicts failures, reduces downtime, and improves energy efficiency for Precision Manufacturing Inc.

Response

The company is experiencing equipment downtime, machine overheating, excessive vibration, and inefficient power consumption. These problems reduce productivity, increase maintenance costs, and affect overall operational efficiency.

Data Collection Needs

The IoT system must collect the following data:

- Temperature data to detect overheating.
- Vibration data to identify component wear and potential failures.
- Power consumption data to monitor energy usage.
- Operational uptime data to measure machine performance and availability.

Automated Actions

The system should automatically:

- Send alerts when temperature exceeds safe limits.
- Send alerts when abnormal vibration is detected.
- Schedule preventive maintenance before equipment failure occurs.
- Reduce power consumption by shutting down or slowing idle machines.
- Generate maintenance reports for engineers.

Constraints

- Security of machine and operational data.
- Reliable network connectivity within the factory.

- Cost-effective deployment and maintenance.
- Scalability for future factory expansion.
- Real-time monitoring requirements with minimal latency.

2. Selection of Sensors and Edge Devices

Response

Sensors

1. Thermocouple Temperature Sensor
    - Detects overheating in industrial machines.
2. Accelerometer (Vibration Sensor)
    - Monitors vibration levels and identifies abnormal machine behavior.
3. Current Sensor
    - Measures machine power consumption for energy optimization.

Actuators

1. Industrial Relay
    - Controls machine power automatically.
    - Helps reduce energy consumption when machines are idle.

Edge Devices

1. ESP32 Microcontroller
    - Collects sensor data.
    - Sends data to the gateway device.
2. Industrial Raspberry Pi or NVIDIA Jetson Nano
    - Processes sensor data locally.
    - Performs edge analytics and anomaly detection.
    - Acts as the IIoT gateway.

These devices were selected because they provide reliable data collection, local processing, and cloud connectivity at a reasonable cost.

3. Connectivity and Communication Needs

Response

The proposed solution uses both local edge processing and cloud-based services.

Communication Protocols

- Wi-Fi for real-time connectivity.
- MQTT protocol for lightweight and secure communication between devices and the cloud.
- LoRaWAN (optional) for long-range communication across large factory floors.

Bandwidth and Power Considerations

MQTT minimizes bandwidth consumption by transmitting small data packets efficiently.

Edge processing reduces the amount of data sent to the cloud, lowering network traffic and improving response times.

LoRaWAN can be used where long-distance communication is needed while maintaining low power consumption.

4. Processing and Storage Options

Response

Processing Method

The system uses a hybrid approach:

Edge Computing

- Raspberry Pi or Jetson Nano processes real-time sensor data.
- Detects anomalies immediately.
- Generates instant alerts.

Cloud Computing

- AWS IoT Greengrass or Azure IoT Edge stores historical data.
- Performs predictive maintenance analytics.

Storage Method

Data will be stored in:

- PostgreSQL for structured machine performance records.

- AWS DynamoDB for scalable cloud storage of sensor data.

This approach provides fast decision-making at the edge while supporting long-term analysis in the cloud.

5. Security Measures Implementation

Response

Data Protection Methods

- AES-256 Encryption protects stored machine data.
- TLS/SSL Encryption secures data transmitted between devices and cloud services.

Access Control Strategy

- Role-Based Access Control (RBAC) limits access to authorized personnel.
- Multi-Factor Authentication (MFA) provides an additional layer of security for engineers and administrators.

System Updates and Maintenance

- Firmware Over-the-Air (FOTA) updates enable remote deployment of security patches.
- Regular vulnerability assessments and security reviews will be conducted.

These controls ensure confidentiality, integrity, and availability of industrial data.

6. User Interface Design

Response

Interface Type

A web-based dashboard developed using React.js will allow engineers and managers to monitor machines remotely.

Dashboard Features

- Real-time machine status monitoring.
- Temperature and vibration trend charts.

- Power consumption analytics.
- Predictive maintenance alerts.
- Operational uptime reports.
- Historical performance records.
- Equipment health score indicators.

User Experience Considerations

The interface should be:

- Easy to navigate.
- Responsive on desktop and mobile devices.
- Designed with clear charts and alerts.
- Capable of displaying critical information at a glance.

7. Final System Architecture Summary

Response

System Architecture Diagram

Sensors → ESP32 → Raspberry Pi/Jetson Nano → MQTT → Cloud Platform → Database → Dashboard

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Industrial Relay (Automated Control)

Component Integration

1. Sensors collect temperature, vibration, power consumption, and uptime data.
2. ESP32 gathers sensor readings.
3. Raspberry Pi or Jetson Nano performs edge processing and anomaly detection.
4. MQTT securely transmits data to the cloud.
5. AWS IoT Greengrass or Azure IoT Edge processes and stores information.
6. PostgreSQL or DynamoDB stores historical records.
7. The dashboard displays real-time analytics and alerts.
8. Industrial relays perform automated energy-saving actions.

Potential Improvements

- AI-powered predictive maintenance models.
- Integration with digital twins.
- Machine learning-based anomaly detection.
- Mobile application support.
- Advanced reporting and forecasting.

8. Conclusion

Response

This IIoT solution provides a comprehensive approach to monitoring industrial equipment and improving operational efficiency. Key design considerations included real-time monitoring, predictive maintenance, energy optimization, security, and scalability.

One challenge was balancing real-time processing requirements with cloud-based analytics while maintaining strong security controls. Another consideration was selecting communication technologies that provide reliable connectivity in industrial environments.

Future improvements could include advanced machine learning models, digital twin technology, and enhanced predictive analytics. Overall, the proposed architecture is practical, secure, scalable, and capable of significantly reducing downtime while improving machine performance and energy efficiency.